Inference HW by Caleb Christian

1.) $SE = \frac{10}{\sqrt{100}} = 1$ $\quad z = \frac{52-50}{1} = 2$

$CI = 52 \pm 1.96(1)$

$= [50.04, 53.96]$

This is the confidence interval for 95%

2.) $\hat{S} = 3.1$, $\widehat{SE}(\hat{S}) = 1.3$

$z = \frac{3.1}{1.3} = 2.4$

Confidence level		CI
90%	$3.1 \pm 1.645(1.3)$	$(0.9615, 5.2385)$ Reject H_0
95%	$3.1 \pm 1.96(1.3)$	$(0.552, 5.648)$ Reject H_0
99%	$3.1 \pm 2.576(1.3)$	$(-0.2488, 6.4488)$ Fail to reject H_0

90% & 95% we can reject H_0 but not 99% because it has an interval containing 0

3) $\bar{x} = 11.5$, $s = 3$, $n = 36$

$$SE = \frac{3}{\sqrt{36}} = \frac{1}{2} \quad \therefore CI = 11.5 \pm 1.96(0.5)$$
$$= 11.5 \pm 0.98$$
$$= [10.52, 12.48]$$

$$z = \frac{11.5 - 12}{0.5} = -1$$

$$p = 2P(Z \leq -1)$$
$$= 2(0.1587)$$
$$= 0.317$$

This is greater than 0.05 so we fail to reject H_0 so the mean μ could be 12 yes

4.) $\bar{x}_A = 25$, $\bar{x}_B = 22$

$25 - 22 = 3$

$$SE = \sqrt{\frac{8^2}{106} + \frac{7^2}{106}} = \sqrt{1.13} = 1.06$$

$$z = \frac{3}{1.06} \approx 2.8$$

$$CI_{95} = 3 \pm 1.96(1.06)$$
$$\approx [0.917, 5.084]$$

5.) if $\bar{x}_A = 14$, $\bar{x}_B = 12$

$$14 - 12 = 2$$

$$SE = \sqrt{\frac{6^2}{81} + \frac{5^2}{64}} = \sqrt{0.44 + 0.39063}$$
$$\approx 0.914$$

$$\therefore z = \frac{2}{0.914} \approx 2.19$$

$$p(z) \approx 0.0286$$

6) $\bar{x}_A = 3.2$, $\bar{x}_B = 2.8$

$$3.2 - 2.8 = 0.4$$

$$SE = \sqrt{\frac{1.5^2}{49} + \frac{1.4^2}{64}} \approx 0.2767$$

$$\therefore z = \frac{0.4}{0.2767} \approx 1.446$$

$$p = 2P(-|1.446|)$$
$$\approx 0.1483$$

$0.1483 > 0.05$ so we fail to reject H_0

We can say that the sample mean diff of 0.4 isn't significant.

Rest of questions are in ipynb